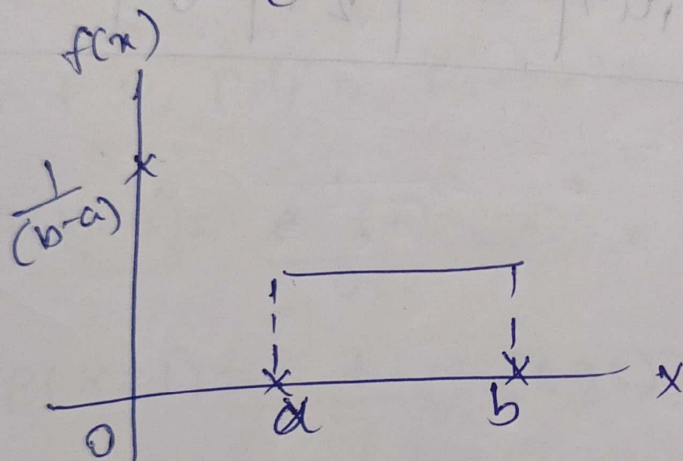


Unit-3

Uniform distribution:

Let x be a continuous random variable the probability density function of uniform distribution given by

$$f(x) = \begin{cases} \frac{1}{b-a} ; a \leq x \leq b \\ 0, \text{ otherwise} \end{cases}$$



$$\text{Mean } \mu = \frac{a+b}{2}$$

$$\text{Variance } \sigma^2 = \frac{(b-a)^2}{12}$$

Find mean and Variance of uniform distribution in the interval $(-3, 2)$

$$a = -3, b = 2 \quad (a, b)$$

$$\text{Mean } \mu = \frac{a+b}{2} = \frac{-3+2}{2} = \frac{-1}{2} = -0.5$$

$$\text{Variance } = \sigma^2 = \frac{(b-a)^2}{12} = \frac{(2+3)^2}{12} = \frac{25}{12} = 2.0833...$$

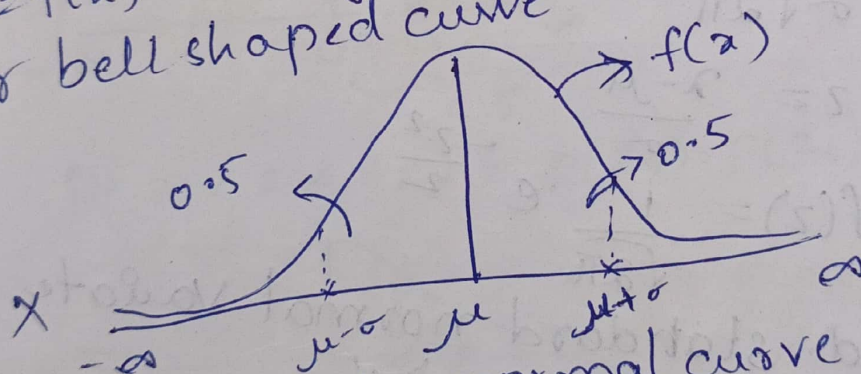
Normal distribution:

Let x be a continuous random variable the probability density function of normal distribution is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad \begin{matrix} -\infty < x < \infty \\ -\infty < \mu < \infty \\ \sigma > 0 \end{matrix}$$

where μ is the mean and σ is the standard deviation. x is called normal variate characteristic of normal distribution.

i) The graph of normal (curve) distribution $y = f(x)$ in x, y plane is called normal curve or bell shaped curve.



ii) Total area under normal curve is unity.

since $\int_{-\infty}^{\infty} f(x) dx = 1$

- iii) The normal curve is symmetric about mean
- iv) The normal curve is maximum at its mean
- v) In normal distribution mean = median = mode
- vi) it is uni model.
- vii) The points where the curve drops are called points of inflexion and are situated at $\mu \pm \sigma$

68.27% area and a normal curve lies between $\mu \pm \sigma$

9) 95.43% area lies between $\mu \pm 2\sigma$

10) 99.73% area lies between $\mu \pm 3\sigma$

$$P(x_1 \leq x \leq x_2) = \frac{1}{\sigma\sqrt{2\pi}} \int_{x_1}^{x_2} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

Standard normal distribution:

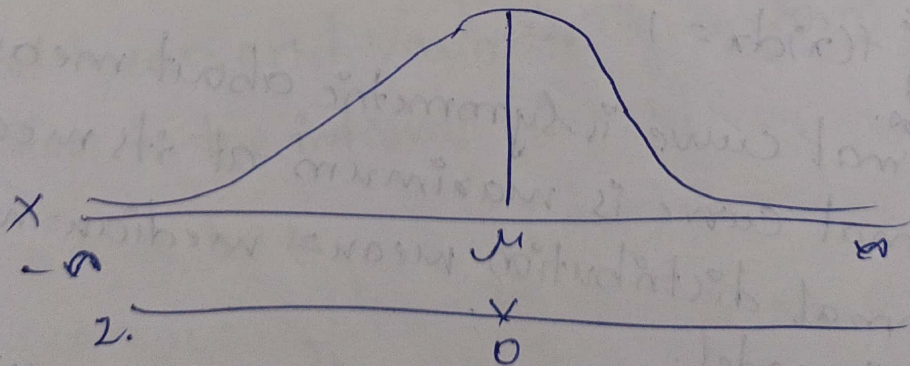
The probability distribution density function in normal distribution is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$$\text{Let } z = \frac{x-\mu}{\sigma}$$

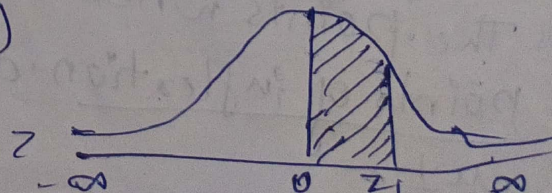
$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$$

z is called standard normal variate with mean zero and $\sigma = 1$

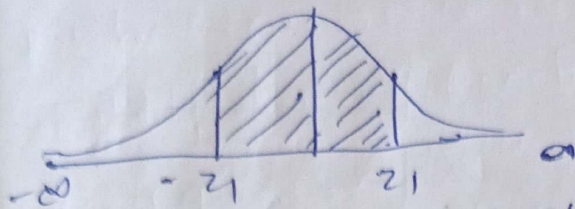


Areas under normal curve.

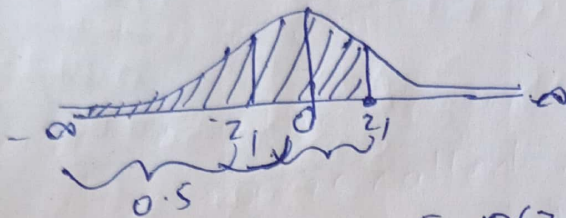
1. $P(0 < z < z_1) = P(z_1)$



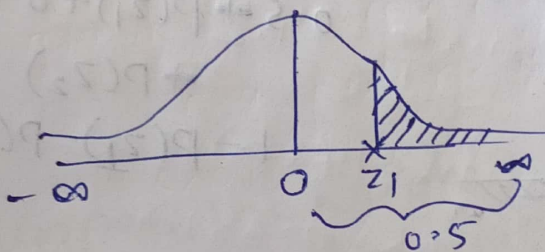
$$2. P(-z_1 < Z < 0) = P(z_1) = P(z_1)$$



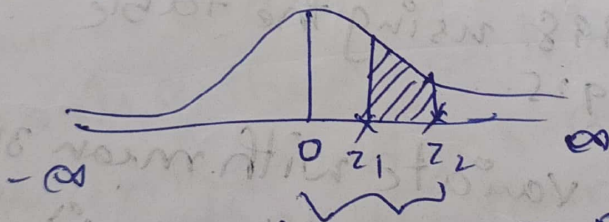
$$3. P(Z < z_1) = 0.5 + P(z_1)$$



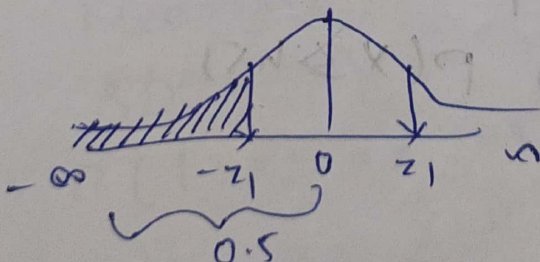
$$4. P(Z > z_1) = 0.5 - P(z_1)$$



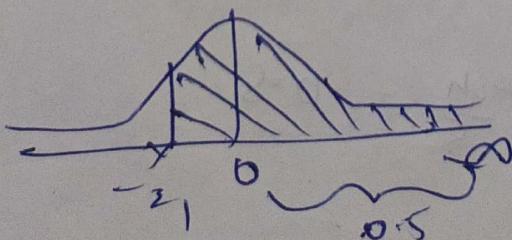
$$5. P(z_1 < Z < z_2) = P(z_2) - P(z_1)$$



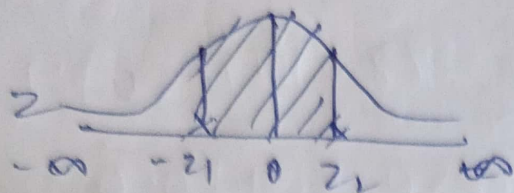
$$6. P(Z < -z_1) = 0.5 - P(z_1)$$



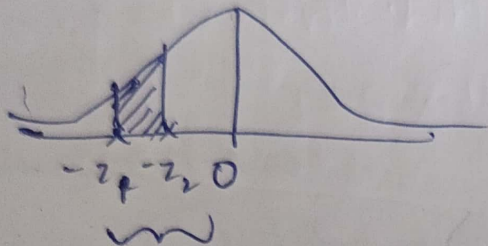
$$7. P(Z > -z_1) = 0.5 + P(z_1)$$



$$\textcircled{8} \quad P(-z_1 < Z < z_2) = P(z_1) + P(z_2)$$



$$\textcircled{9} \quad P(-z_1 < Z < z_2) = P(z_1) - P(z_2)$$

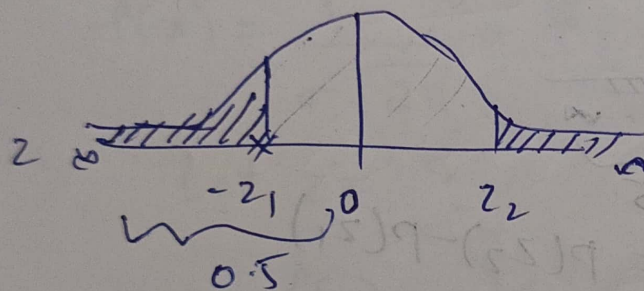


$$\textcircled{10} \quad P(Z < -z_1 \text{ and } Z > z_2) = P(Z < -z_1) + P(Z > z_2)$$

$$0.5 - P(z_1) + 0.5$$

$$+ P(z_2)$$

$$1 - P(z_1) - P(z_2)$$



Ex: $P(3.57) = 0.4998$ using the table
 $P(1.24) = 0.3925$

Q: X is a normal variate with mean 30 and s.d. 5. Find probability of i)

$$P(26 \leq X \leq 40) \quad \text{ii) } P(X \geq 45)$$

$$\mu = 30, \sigma = 5$$

$$\text{(i)} \quad Z = \frac{X - \mu}{\sigma}$$

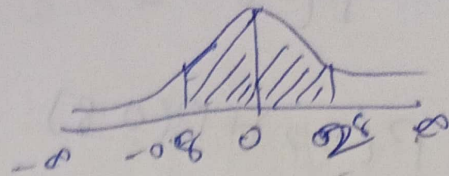
$$Z_1 = \frac{26 - 30}{5} = -\frac{4}{5} = -0.8$$

$$z_2 = \frac{40-30}{5} = \frac{10}{5} = 2$$

$$P(-0.8 \leq z \leq 2)$$

$$= P(-0.8) + P(2)$$

$$= 0.2881 + 0.4772 = 0.7653$$



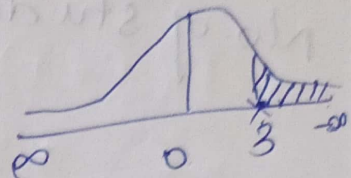
$$ii) P(X \geq 25)$$

$$z_1 = \frac{25-30}{5} = -1$$

$$P(Z \geq 3) = 0.5 - P(3)$$

$$= 0.5 - 0.4987$$

$$= 0.0013$$



Q) Suppose the weight of 800 male students are normally distributed with mean 140 pounds and S.D 10 pounds. Find the marks of the students whose weight lies between 138 and 148 pounds more than 152 pounds.

$$eg \text{ Mean } \mu = 140$$

$$SD \cdot \sigma = 10$$

$$(i) P(138 \leq X \leq 148)$$

$$\text{We've } z = \frac{x - \mu}{\sigma}$$

$$z_1 = \frac{138 - 140}{10} = -0.2$$

$$z_2 = \frac{148 - 140}{10} = 0.8$$

$$P(-0.2 \leq z \leq 0.8)$$

$$= P(0.2) + P(0.8)$$

$$= 0.0793 + 0.2881$$

$$= 0.3674$$

No. of student whose wt lies b/w

$$138 \times 148 = N \times P$$

$$= 800 \times 0.3674$$

$$= 294$$

$$Q1) P(X \geq 152)$$

$$Z_1 = \frac{152 - 140}{10} = \frac{12}{10} = 1.2$$

$$P(Z \geq 1.2) = 0.5 - P(1.2)$$

$$= 0.5 - 0.3849$$

$$= 0.1151$$

No. of Students whose wt greater than 152

$$= N \times P$$

$$= 800 \times 0.1151 = 92$$

In a sample of 1000 cases the mean of certain test is 14 and S.D 2.5. Find how many

students scored (i) b/w 12 and 15

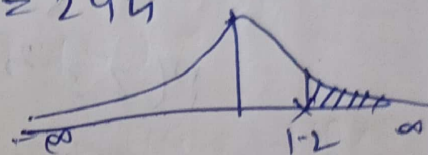
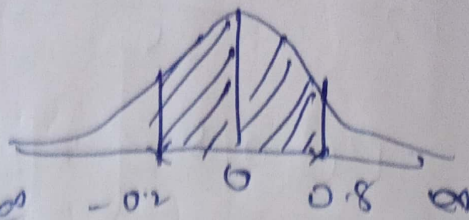
ii) above 18 ii) below 18

$$\text{Mean} = \mu = 14$$

$$\sigma = 2.5$$

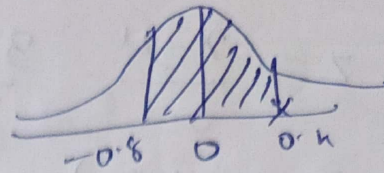
$$(i) P(12 \leq z \leq 15)$$

$$Z_1 = \frac{x - \mu}{\sigma} = \frac{12 - 14}{2.5} = \frac{-2}{2.5} = -0.8$$



$$z_2 = \frac{2 - \mu}{\sigma} = \frac{15 - 14}{2.5} = \frac{1}{2.5} = 0.4$$

$$\begin{aligned} P(-0.8 \leq Z \leq 0.4) \\ &= P(-0.8) + P(0.4) \\ &= 0.2881 + 0.1554 \\ &= 0.4435 \end{aligned}$$



$$\begin{aligned} \text{No. of students} &= N \times 0.4435 \\ &= 444 \end{aligned}$$

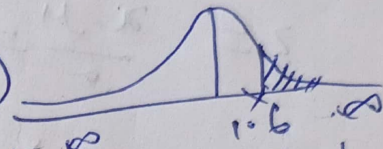
ii) $P(X \geq 18)$

$$z = \frac{18 - 14}{2.5} = \frac{4}{2.5} = 1.6$$

$$P(Z \geq 1.6) = 0.5 - P(1.6)$$

$$= 0.5 - 0.4452$$

$$= 0.0548 \times \text{No. of students} = 0.0548 \times 1000 = 55$$



iii) $P(X \leq 18)$

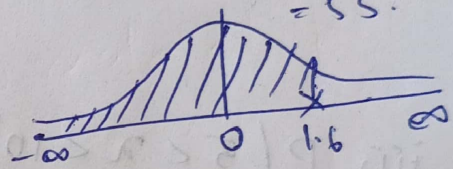
$$z = \frac{18 - 14}{2.5} = 1.6$$

$$P(Z \leq 1.6) = 0.5 + P(1.6)$$

$$= 0.5 + 0.4452$$

$$= 0.9452$$

$$\begin{aligned} \text{No. of students} &= N \times P \\ &= 1000 \times 0.9452 \\ &= 945 \end{aligned}$$

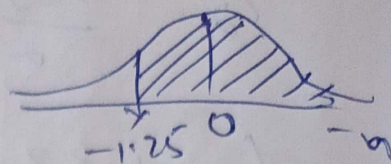


Suppose 2% of people on average of left handed find the probability i) 3 or more ii) ≤ 1 iii) between 5 and 10 iv) > 5 if mean of SD are 8 and 4 respectively.

$$\begin{aligned} \text{Mean} &= \mu = 8 \\ \text{S.D} &= \sigma = 4 \end{aligned}$$

i) $P(x > 3)$

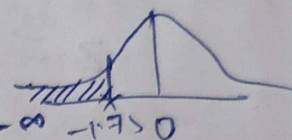
$$z = \frac{x - \mu}{\sigma} = \frac{3 - 8}{4} = -1.25$$



$$P(z > -1.25) = 0.5 + P(-1.25) \\ = 0.5 + 0.3944 = 0.8944 \\ = 0.8944 \times 100\% \\ = 89.44\%$$

ii) $P(x \leq 1)$

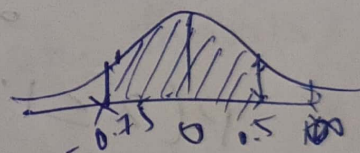
$$z = \frac{x - \mu}{\sigma} = \frac{1 - 8}{4} = -1.75$$



$$P(z \leq -1.75) = 0.5 + P(-1.75) \\ = 0.5 + 0.4599 = 0.9599 \\ = 0.9599 \times 100\% \\ = 95.99\%$$

iii) $P(5 < x < 10)$

$$z_1 = \frac{x - \mu}{\sigma} = \frac{5 - 8}{4} = -0.75$$



$$z_2 = \frac{10 - 8}{4} = \frac{2}{4} = 0.5$$

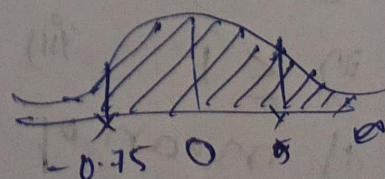
$$P(-0.75 \leq z \leq 0.5)$$

$$= P(-0.75) + P(0.5)$$

$$= 0.2734 + 0.1915 = 0.4649$$

iv) $P(x > 5)$

$$z = \frac{5 - 8}{4} = -0.75$$



$$P(z > -0.75)$$

$$= 0.5 + P(-0.71)$$

$$= 0.5 + 0.2234$$

$$= 0.7234$$

✓) X is normally distributed with mean 2 variance 0.1. Find the probability of

$$P(|X-2| > 0.01)$$

$$\text{Mean } \mu = 2$$

$$\text{SD} = \sigma = \sqrt{0.1} = 0.316$$

$$\text{Wkt, } P(|x - \mu| < \alpha)$$

$$= P(\mu - \alpha < x < \mu + \alpha)$$

$$\text{Given } P(|x - 2| > 0.01)$$

$$= 1 - P(|x - 2| < 0.01)$$

$$= 1 - P(2 - 0.01 < x < 2 + 0.01)$$

$$= 1 - P(1.99 < x < 2.01)$$

$$= 1 - P(1.99 < \underline{x} < 2.01)$$

$$P(0.19 < x < 2.01)$$

$$z_1 = \frac{1.99 - 2}{0.316} = -0.0316$$

$$= -0.0120$$

$$z_2 = \frac{2.01 - 2}{0.316} = 0.0316$$

$$P(-0.0316 < z < 0.0316)$$

$$= P(-0.0316) + P(0.0316)$$

$$= 0.0120 + 0.0120 = 0.024$$

$$1 - 0.024 = 0.976$$

$$0.4778$$

$$0.4778$$

$$0.0044$$



Normal Approximation to Binomial Distribution:

For large 'n' we can approximate the Binomial curve by normal curve and calculate the probability for a no. of success the real class interval is $(x_1 - 0.5, x_2 + 0.5)$

Here we take $z_1 = \frac{(x_1 - 0.5) - np}{\sqrt{npq}}$,

$$z_2 = \frac{(x_2 + 0.5) - np}{\sqrt{npq}}.$$

1. Find the probability that out of 100 patients b/w 84 and 95 inclusive will survive a heart operation. Given the chance of survival is 0.9

$$n = 100$$

$$p = 0.9$$

$$q = 1 - p = 0.1$$

$$\text{Mean} = \mu = np = (100)(0.9) = 90$$

$$\text{Variance} = \mu = npq = (100)(0.9)(0.1) = 9$$

$$\text{Std-dev} = \sqrt{npq} = \sqrt{9} = 3.$$

$$P(84 \leq x \leq 95)$$

Using normal approximation to binomial distribution we calculate probability of

$$P(83.5 \leq x \leq 95.5)$$

$$z_1 = \frac{(x_1 - 0.5) - np}{\sqrt{npq}} = \frac{(83.5 - 90)}{3}$$

$$z = \frac{-13}{6} = -2.166$$

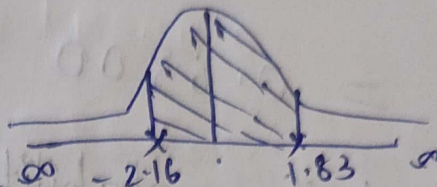
$$z_2 = \frac{(x_2 + 0.5) - np}{\sqrt{npq}} = \frac{95.5 - 90}{3} = \frac{5.5}{3} = 1.8333$$

$$p(-2.16 < z < 1.83)$$

$$= p(2.16) + p(1.83)$$

$$= 0.4846 + 0.4664$$

$$= 0.951$$



8 coins are tossed together find the probability of getting 1 to 4 heads in a single toss.

$$n = 8$$

$$p = \frac{1}{2} \text{ (prob of head)}$$

$$q = \frac{1}{2}$$

$$\text{Mean } = \mu = np = 8\left(\frac{1}{2}\right) = 4$$

$$\text{Variance} = npq = 8\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = 2$$

$$\text{Std. dev} = \sqrt{2} = 1.414$$

$$p(1 \leq z \leq 4)$$

Using normal approximation to binomial distribution

$$p(0.5 \leq z \leq 4.5)$$

$$z_1 = \frac{(x_1 - 0.5) - np}{\sqrt{npq}} = \frac{0.5 - 4}{1.414}$$

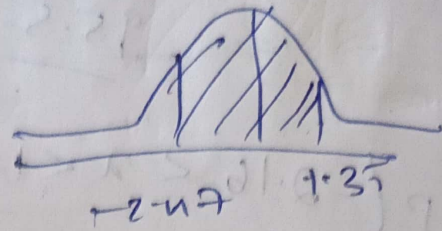
$$= -2.47$$

$$= -2.47$$

$$z_2 = \frac{0.5 - 0}{1.414} = \frac{0.5}{1.414} = 0.35$$

$$P(-2.47 \leq z \leq 0.35) \\ = P(2.47) + P(0.35) \\ = 0.4932 + 0.1368$$

$$= 0.6300$$



Find the probability that by guess work a student can correctly answer 25-30 questions in a MCQ quiz consisting of 80 questions assume that in each question with 4 choices only one choice is correct and student has no knowledge of the subject

$$n = 80$$

$$p = \frac{1}{4} \text{ (prob of correct answer)}$$

$$q = \frac{3}{4}$$

$$\text{Mean} = \mu = np = 80\left(\frac{1}{4}\right) = 20$$

$$\text{Variance} = \sigma^2 = npq = 80\left(\frac{1}{4}\right)\left(\frac{3}{4}\right) = 15$$

$$\text{std. deviation} = \sqrt{15} = 3.8729$$

Using normal approximation

$$x_1 = 25; x_2 = 30$$

$$P(24.5 \leq x \leq 30.5)$$

$$z_1 = \frac{(x_1 - 0.5) - np}{\sqrt{npq}} = \frac{24.5 - 20}{3.872} = \frac{4.5}{3.872} = 1.16$$

$$z_2 = \frac{(x_2 + 0.5) - np}{\sqrt{npq}}$$

$$= \frac{(30.5) - 20}{3.872} = \frac{10.5}{3.872}$$

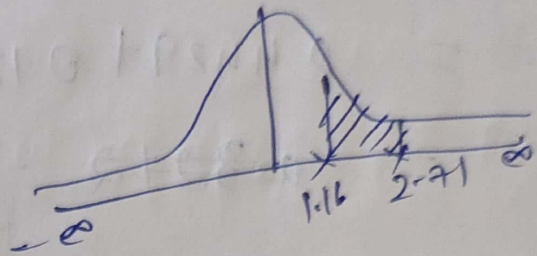
$$= 2.71$$

$$P(1.16 \leq z \leq 2.71)$$

$$P(2.71) - P(1.16)$$

$$= 0.4966 - 0.3770$$

$$= 0.1196$$



① determine the probability that getting an even number on the face 3 to 5 times in throwing 10 dice together.

$$n = 10$$

$$p = \frac{1}{2} \text{ (prob of even)}.$$

$$q = \frac{1}{2}$$

$$\text{Mean} = \mu = np = 10\left(\frac{1}{2}\right) = 5$$

$$\text{Var} = \sigma^2 = npq = 10\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = 2.5$$

$$\sigma = \sqrt{2.5} = 1.581$$

Using normal approximation

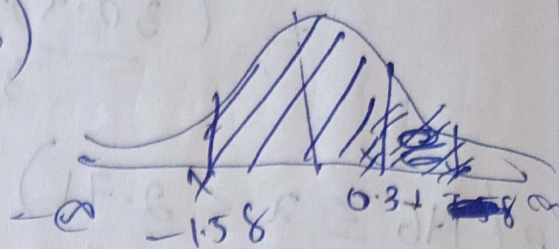
$$x_1 = 3 \quad x_2 = 5$$

$$P(2.5 \leq z \leq 5.5)$$

$$z_1 = \frac{2.5 - 5}{1.581} = \frac{-2.5}{1.581} = -1.58$$

$$z_2 = \frac{5.5 - 5}{1.581} = \frac{0.5}{1.581} \approx 0.31$$

$$\begin{aligned} & P(0.31 \leq z \leq -1.58) \\ &= P(1.58) + P(0.31) \\ &= 0.4429 + 0.1217 \\ &= \cancel{0.3212} = 0.5646 \end{aligned}$$



Population: It is the set of all items under consideration.

The no. of items in the population is called population size and is denoted by 'N'.

The measurement of the population are called Parameters. They are.

Population mean = μ .

population variance = σ^2

Population std. dev = σ

population proportion = p .

Sample: It is the subset of population.

The no. of items in the sample is called Sample Size. It is denoted by 'n'.

If $n \geq 30 \rightarrow$ It is called sample.

If $n < 30 \rightarrow$ It is called small sample.

The mathematical measurements of the sample are called statistics. They are.

Sample mean $= \bar{x} = \frac{\sum x_i}{n}$

sample variance $= s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$

sample st d. dev $= s$

sample Proportion $\bar{p} = \frac{x}{n}$

Random sampling:

The process of drawing sample from the population in such a way that each member of the population of being included in the sample is called random sampling.

If 'N' is the population size and 'n' is the sample size:

1) If no. of samples drawn from the population without replacement $= N C n$.

2) The no. of samples drawn from the population with replacement $= N^n$

Sampling distribution:

It is a distribution of the statistic which is the frequency distribution formed with various values of a statistic like mean, variance etc.

computed from different samples of same size from a population of fixed size.

Standard error of a statistics (S.E.),

The standard error of a statistics is the

standard deviation of the sampling distribution of statistics.

Formulas:

Let N be the Population size and n be the Sample size

(i) Standard error of mean (infinite Population)
(SE) $= \frac{\sigma}{\sqrt{n}}$

(ii) SE of mean (finite population)
 $= \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$ $\rightarrow$ finite population correction factor.

(iii) SE of Proportion (Infinite population) $= \sqrt{\frac{PQ}{n}}$,
($Q = 1 - P$)

(iv) SE of Proportion (finite population) $= \sqrt{\frac{PQ}{n}} \sqrt{\frac{N-n}{N-1}}$

SE for the difference between two means.

$$SE \text{ of } (\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

(v) SE for difference between Sample proportions

$$SE \text{ of } (\bar{P}_1 - \bar{P}_2) = \sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}$$

Sampling distribution of means: (σ - known)

1. What is the value of correction factor if $n = 5$

and $N = 200$ (finite population correction factor)

$$= \left(\sqrt{\frac{N-n}{N-1}} \right)$$

$$= \left(\sqrt{\frac{200-5}{200-1}} \right) = \left(\sqrt{\frac{195}{199}} \right) = 0.9898$$

$$0.9798$$

1. population consists of 5 numbers 2, 3, 6, 8, 11

consider all possible samples of size 2

a) without replacement

b) with replacement and find a) Mean of Population

b) Std. dev of Population c) Mean of sampling distribution of means d) Std. dev of sampling distribution of means.

2	3	6	
	8	11	

Population size $N = 5$

a) Population mean $= \mu = \frac{\sum x_i}{N} = \frac{2+3+6+8+11}{5} = \frac{30}{5} = 6$

Population variance $= \sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$

$= \frac{(2-6)^2 + (3-6)^2 + (6-6)^2 + (8-6)^2 + (11-6)^2}{5}$
 $= \frac{16 + 9 + 0 + 4 + 25}{5} = \frac{54}{5} = 10.8$

Population Std. dev $= \sigma = \sqrt{\sigma^2} = \sqrt{10.8} = 3.286$

(i) Without replacement:

Sample size $= n = 2$

No. of samples drawn from the population

without replacement $= Nc_n = 5c_2 = 10$

(2, 3)
2.5

(2, 6)
4

(2, 8)
5

(2, 11)
6.5

(3, 6)
4.5

(3, 8)
5.5

(3, 11)
7

(6, 8)
7

(6, 11)
8.5

(8, 11)
9.5

Sampling distribution of means

c) Mean of sampling dist of mean = $\mu_{\bar{x}}$

$$= \frac{2.5 + 4 + 5 + 6.5 + 4.5 + 5.5 + 7 + 7 + 8.5 + 9.5}{10}$$

$$= \frac{60}{10} = 6$$

$$\boxed{\mu_{\bar{x}} = \mu}$$

d) Std. dev of sampling dist of mean = $\sigma_{\bar{x}}$

variance of sampling dist of mean = $\sigma_{\bar{x}}^2$

$$= \frac{\sum (x_i - \mu_{\bar{x}})^2}{10}$$

$$= \frac{(2.5-6)^2 + (4-6)^2 + (6.5-6)^2 + (4.5-6)^2 + (5.5-6)^2 + (7-6)^2 + (7-6)^2 + (8.5-6)^2 + (9.5-6)^2 + (5-6)^2}{10}$$

$$= \frac{12.25 + 4 + 0.25 + 2.25 + 0.25 + 1 + 1 + 6.25 + 12.25 + 1}{10}$$

$$= 3.95 + 1 = 4.95$$

Std. dev of sampling dist of mean

$$\sigma_{\bar{x}} = \sqrt{4.95} = 2.01$$

(ii) With Replacement:

No. of samples of size 2 that can be drawn with replacement = $N^n = 5^2 = 25$

(2, 2) 2	(2, 3) 2.5	(2, 6) 4	(2, 8) 5	(2, 11) 6.5
(3, 2) 2.5	(3, 3) 3	(3, 6) 4.5	(3, 8) 5.5	(3, 11) 7
(6, 2) 4	(6, 3) 4.5	(6, 6) 6	(6, 8) 7	(6, 11) 8.5
(8, 2) 5	(8, 3) 5.5	(8, 6) 7	(8, 8) 8	(8, 11) 9.5
(11, 2) 6.5	(11, 3) 7	(11, 6) 8.5	(11, 8) 9.5	(11, 11) 11

sampling distribution of mean.

$$= \frac{2 + 2.5 + \dots + 11}{25}$$

$$= \frac{150}{25} = 6$$

$$\boxed{\mu_{\bar{x}} = \mu}$$

std. dev of sampling dist of mean
variance of sampling dist of mean = $\sigma^2 / 2$

$$= \frac{\sum (x_i - \mu_{\bar{x}})^2}{25}$$

$$= \frac{(2-6)^2 + (2.5-6)^2 + (4-6)^2 + \dots + (11-6)^2}{25}$$

$$= 16 + 12.25 + 4 + 1 + 0.25 + 12.25 + 9 + 2.25 + 0.25 + 1 + 4 + 2.25 + 0 + 1 + 6.25 + 1 + 4 + 12.25 + 0.25 + 1 + 6.25 + 12.25 + 25 = 135$$

$$\frac{135}{25} = 5.4$$

Std. dev of sampling dist of mean

$$= \sqrt{5.398} = \sqrt{5.4}$$

$$= 2.323 = 2.323$$

Q A population consists of 6 mem n, 8, 12, 16, 20, 24. Consider all possible samples of size 2 i) without replacement ii) with replacement.

a) mean of population

b) Standard deviation

c) mean of sampling distribution of means

d) SD

Population $N = 6$

n	8	12
16	20	24

a) Population mean = $\mu = \frac{\sum x_i}{N}$

$$= \frac{4 + 8 + 12 + 16 + 20 + 24}{6} = \frac{84}{6} = 14$$

Population variance = $\frac{\sum (x_i - \mu)^2}{N}$

$$= \frac{(4-14)^2 + (8-14)^2 + (12-14)^2 + (16-14)^2 + (20-14)^2 + (24-14)^2}{6} = \frac{100 + 36 + 4 + 4 + 36 + 100}{6}$$

$$= \frac{280}{61} \approx 4.6 \cdot 6$$

$$= 46.66$$

$$\text{std. dev} = \sigma = \sqrt{\sigma^2} = \sqrt{46.66} = 6.831$$

without replacement

sample size $n = 2$

No. of samples of size 2 drawn from the population

without replacement: $N C n = 6 C 2 = 15$

$$(4, 8) \quad (4, 12) \quad (4, 16) \quad (4, 20) \quad \left| \begin{array}{c} 4 \quad 8 \quad 12 \\ 16 \quad 20 \quad 24 \end{array} \right.$$

6 8 10 12

$$(4, 24) \quad (8, 12) \quad (8, 16) \quad (8, 20) \quad (8, 24)$$

14 10 12 14 16

$$(12, 16) \quad (12, 20) \quad (12, 24) \quad (16, 20)$$

14 16 18 18

$$(16, 24) \quad (20, 24)$$

20 22

Mean of sampling dist of mean $= \mu_{\bar{x}}$

$$= \frac{6 + 8 + 10 + 12 + 14 + 10 + 12 + 14 + 16 + 14 + 16 + 18 + 18 + 20 + 22}{15} = \frac{210}{15}$$

$$= 14$$

$$\boxed{\mu_{\bar{x}} = \mu}$$

Variance of sampling dist mean = σ^2/n

$$= \frac{\sum (x_i - \mu)^2}{n}$$

$$= \frac{(6-14)^2 + (8-14)^2 + (10-14)^2 + (12-14)^2 + (14-14)^2 + (10-14)^2 + (12-14)^2 + (14-14)^2 + (16-14)^2 + (14-14)^2 + (18-14)^2 + (10-18)^2 + (20-14)^2 + (22-14)^2}{15}$$

$$= \frac{64 + 36 + 16 + 4 + 0 + 16 + 4 + 0 + 4 + 16 + 16 + 36 + 64 + 36}{15}$$

$$= \frac{280}{15} = 18.666 \dots$$

d) Std. dev of Sampling dist of mean = σ

$$= \sqrt{\sigma^2/n} = \sqrt{18.666} = 4.320$$

ii) With replacement

No. of samples of size 2 that can be drawn

With replacement $= N^n = 6^2 = 36$ 4, 8, 12
16, 20, 24

(4, 4)	(4, 8)	(4, 12)	(4, 16)	(4, 20)
4	6	8	10	12
(4, 24)	(8, 4)	(8, 8)	(8, 12)	(8, 16)
16	6	8	10	12

(8, 20)	(8, 24)	(12, 4)	(12, 8)	(12, 12)
14	16	8	10	12
(12, 16)	(12, 20)	(12, 24)	(16, 4)	(16, 8)
14	16	18	10	12
(16, 12)	(16, 16)	(16, 20)	(16, 24)	(20, 4)
14	16	18	20	12
(20, 8)	(20, 12)	(20, 16)	(20, 20)	(20, 24)
14	16	18	20	22
(24, 4)	(24, 8)	(24, 12)	(24, 16)	(24, 20)
14	16	18	20	22
(24, 24)				

24 : samples
 mean of distribution of mean = $\mu_{\bar{x}}$

$$= \frac{4 + 6 + 8 + 10 + 12 + 14 + 16 + 18 + 20 + 22 + 24}{36} = 14$$

$$= \frac{504}{36} = 14$$

Variance of sampling distribution of mean = $\sigma_{\bar{x}}^2$

$$= \frac{\sum (x_i - \mu_{\bar{x}})^2}{36}$$

$$= 100 + 64 + 36 + 16 + 4 + 0 + 64 + 36 + 16 + 4 + 0 + 4 + 16 + 16 + 4 + 0 + 4 + 36 + 16 + 4 + 0 + 4 + 16 + 36 + 64 + 100 = 504$$

$$= \frac{840}{36} = 23.333 \dots$$

d) Std. dev of sampling dist of mean

$$\sigma = \sqrt{\frac{\sigma^2}{n}} = \sqrt{23.333} = 4.830$$

Central limit theorem:

If $\bar{x}$ be the mean of a random sample of size 'n' ~~from~~ drawn from a population having mean ' μ ' and std. dev ' σ ' then $z = \frac{\bar{x} - \mu}{(\sigma/\sqrt{n})}$

is a random variable whose distribution approaches standard normal distribution as n tends to ∞ ($n \rightarrow \infty$)

① The mean height of students in a college is 155cm and std. dev is 15. What is the probability that the mean height of 36 students is < 157 cm

Population mean $\mu = 155$ cm

" std. dev $\sigma = 15$

Sample size $n = 36$

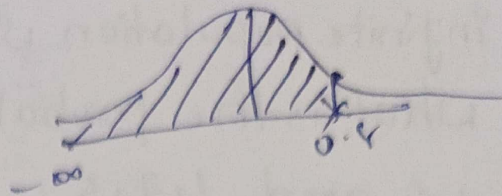
To find

$$P(\bar{x} < 157)$$

Using central limit theorem,

$$z = \frac{\bar{x} - \mu}{(\sigma/\sqrt{n})} = \frac{157 - 155}{(15/\sqrt{36})} = \frac{157 - 155}{2.5} = 0.8$$

$$\begin{aligned}
 P(Z < 0.8) \\
 &= 0.5 + P(0.8) \\
 &= 0.5 + 0.2881 \\
 &= 0.7881.
 \end{aligned}$$



Probability of mean height of 36 students, $< 157\text{cm}$
 $= 0.7881$.

② The mean of infinite population $\mu = 46$ and variance $= 256$ a random sample of 100 is taken from the population what is the probability that $\bar{x}$ will be between 45 and 78

Population mean $\mu = 46$

population std dev $\sigma = \sqrt{256} = 16$

Sample size $n = 100$.

To find

$$P(75 < \bar{x} < 78)$$

$$Z_1 = \frac{\bar{x} - \mu}{(\sigma/\sqrt{n})} = \frac{75 - 46}{(16/10)} = \frac{-29}{1.6} = -18.125$$

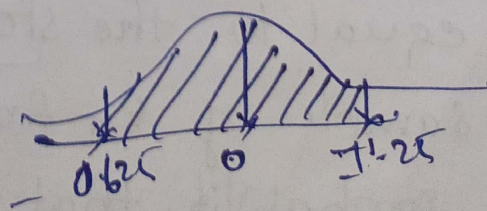
$$Z_2 = \frac{78 - 46}{(16/10)} = \frac{32}{1.6} = 20$$

$$P(-18.125 < Z < 20)$$

$$= P(18.125) + P(20)$$

$$= 0.2324 + 0.3944$$

$$= 0.6268$$



⑧ A random sample of size 64 taken from infinite population with mean $\mu = 45$ and s.d '8'. What is the probability that $\bar{x}$ will be b/w 46 and 47.5.

Population mean = $\mu = 45$

Std. dev = $\sigma = 8$

Sample size $n = 64$

$$P(46 < \bar{x} < 47.5)$$

$$z_1 = \frac{\bar{x} - \mu}{(\sigma/\sqrt{n})} = \frac{46 - 45}{(8/8)} = 1$$

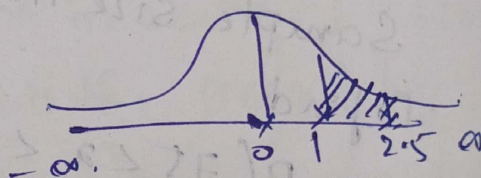
$$z_2 = \frac{47.5 - 45}{1} = 2.5$$

$$P(1 < z < 2.5)$$

$$P(2.5) - P(1)$$

$$= 0.4938 - 0.3413$$

$$= 0.1525$$



Probability of mean height of 64 between 45 and 47.5 is 0.1525

⑨ The mean of certain normal population is equal to the standard error of the mean of the sample of 64 from that distribution. Find the probability that the mean of the sample size 36 will be negative.

$$\text{std. err of mean} = \frac{\sigma}{\sqrt{n}} = \frac{\sigma}{8}$$

$$\text{population mean } \mu = \frac{\sigma}{8}$$

$$\text{sample size } n = 36$$

using central limit theorem;

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$$\text{std. err of mean} = \frac{\bar{x} - \sigma/8}{(\sigma/6)}$$

$$= \frac{6\bar{x} - 0.75}{\sigma}$$

when the $\bar{x}$ will be negative. ($z < -0.75$)

$$z + 0.75 = \frac{6\bar{x}}{\sigma}$$

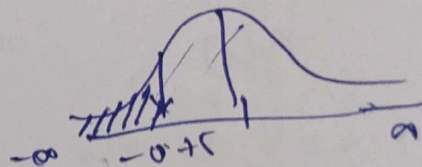
$$\sigma(z + 0.75) = 6\bar{x}$$

$$p(z < -0.75)$$

$$0.5 - p(0.75)$$

$$= 0.5 - 0.2734$$

$$= 0.2266$$



normal population has mean 0.1 and SD 2.1
find the probability that mean of sample size
300 will be negative.

$$\text{population mean } \mu = 0.1$$

$$\sigma = 2.1$$

$$\text{Sample size } n = 300$$

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$$= \frac{\bar{x} - 0.1}{2.1/30} = \frac{\bar{x} - 0.1}{0.07}$$

$$(0.07)Z = \bar{x} - 0.1$$

$$\bar{x} = 0.07Z + 0.1$$

$$Z(0.07) + 0.1 < \bar{x} \quad Z(0.07) + 0.01 < 0$$

$$Z < 1$$

$$\frac{\bar{x} - 0.1}{0.07} < 1$$

$$\bar{x} < 0.07 + 0.1$$

$$\bar{x} < 0.17$$

$$Z(0.07) < -0.1$$

$$Z < \frac{-0.1}{0.07}$$

$$Z < -1.428$$

$$P(Z < -1.428)$$

$$= 0.5 - P(1.428)$$

$$= 0.5 - 0.4222$$

$$= 0.0778$$

When a sample is taken from a infinite population what happens to std-error of mean if the sample size is decreased from 800 to 200.
Standard error of mean

std. error (SE) of mean = $\frac{\sigma}{\sqrt{n}}$

$n_1 = 800$

$SE_1 = \frac{\sigma}{\sqrt{800}} = \frac{\sigma}{20\sqrt{2}}$

$n_2 = 200$

$SE_2 = \frac{\sigma}{\sqrt{200}} = \frac{\sigma}{10\sqrt{2}}$

$SE_1 = \frac{\sigma}{20\sqrt{2}} = \frac{1}{2} \left[\frac{\sigma}{10\sqrt{2}} \right]$

$SE_1 = \frac{1}{2} [SE_2]$

$SE_2 = 2 SE_1$

If the sample size is reduced from 800 to 200 standard error of mean will be multiplied by 2.

Student's t-distribution:

It is used when sample size is small (< 30) and population SD is unknown

The probability density function

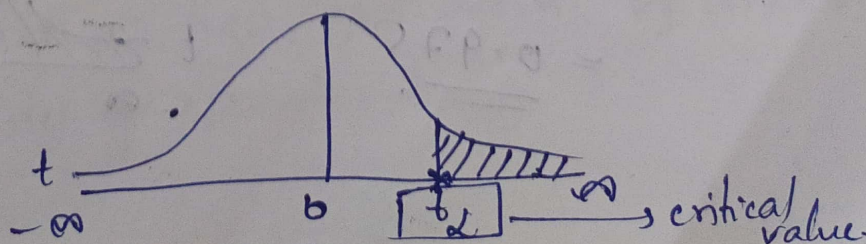
$f(t) = y_0 \left[1 + \frac{t^2}{v} \right]^{-\frac{v+1}{2}}$

where $v = n-1$,
(N.D.)

y_0 is a constant

Properties of t-distribution:

The shape of t-distribution is bell shaped, similar to normal distribution



- > t is symmetric about the line $t=0$
- > t is unimodal

Mean = median = mode

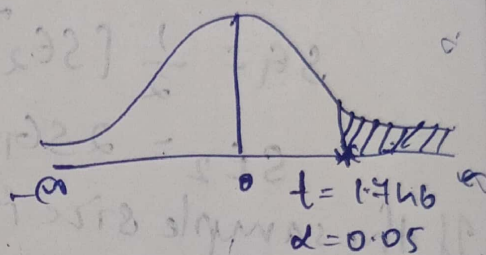
- > The mean of t -distribution is zero and depends upon the parameter $\sqrt{(\text{df})}$ called degree of freedom (dof)

Note: $t_{1-\alpha} = -t_{\alpha}$ (When $\alpha > 0.5$)

α -area
 t_{α} -critical value.

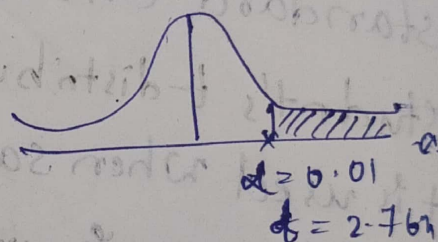
Find $t_{0.05}$ when $\sqrt{=16}$

$$= 1.7546$$



Find $t_{0.01}$ at $\sqrt{10}$

$$= 2.764$$



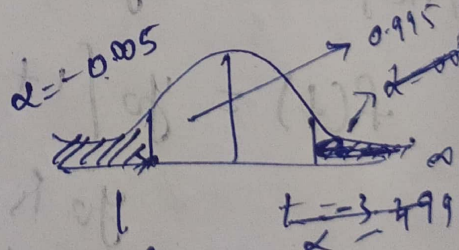
① $t_{0.995}$ at $\sqrt{=7}$

$$t_{1-0.005}$$

$$\Rightarrow -t_{0.005}$$

$$= -3.499$$

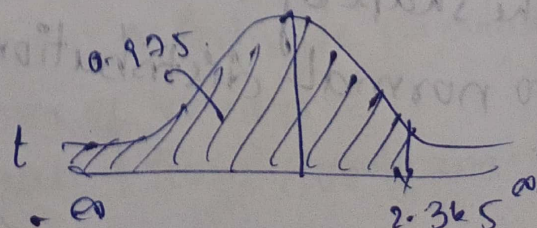
($t_{1-\alpha} = -t_{\alpha}$) ($\alpha > 0.5$)
We will use



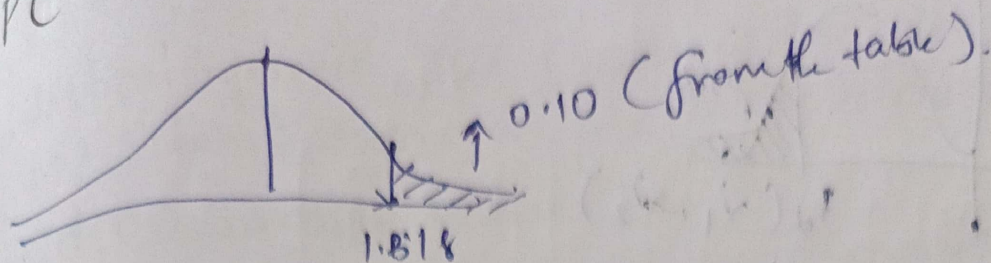
④ Find $p(t < 2.365)$ when $\sqrt{=7}$

$$= 1 - 0.025$$

$$= 0.975$$



① $P(t > 1.318)$ at $\sqrt{24}$.

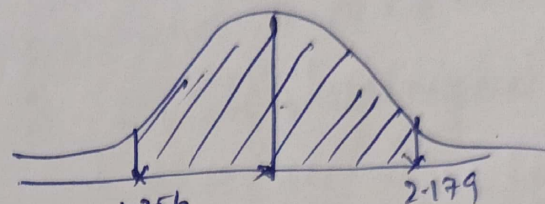
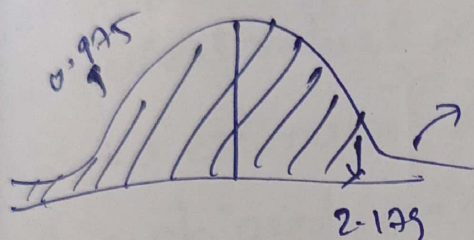


$\alpha = 0.10$

② $P(-1.356 < t < 2.179)$ at $\sqrt{12}$

$P(t < 2.179)$

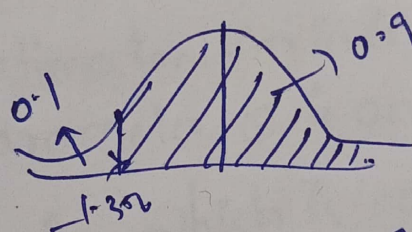
$\sqrt{12} = 0.975$



$P(t > -1.356)$ at $\sqrt{12}$ 0.9

L.A

$$\begin{array}{r} 2.179 \\ -1.356 \\ \hline 3.535 \\ 2.545 \end{array}$$
 (L.A.S. 535)



$P(-1.356 < t < 2.179) = 0.975 - 0.1$
 $= 0.875\%$

F-distribution:

It is used to test the equality of several population means by comparing sample variances.

Properties:

F-distribution curves lie entirely in 1st quadrant.
 F-curve depends on two parameters ν_1 and ν_2 and their order of appearance.

